An ergodic approach to equations of the form $x + y = \lfloor \alpha(n) \rfloor$.

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AI was used to help create these slides, improve the writing,
and produce the figures.

Background

In 1989, Erdős, Sárközy and Sós conjectured that every finite coloring of $\mathbb{N}$ admits distinct monochromatic $x, y \in \mathbb{N}$ and some $n \in \mathbb{N}$ such that

$$x + y = n^2.$$

More generally, they conjectured the same for the equation $x + y = p(n)$, where $p \in \mathbb{Z}[t]$, $p(n) \to +\infty$ and 2 is a prime divisor of p.

When proving this conjecture, the condition "2 is a prime divisor of p" can be reduced to $p(0) = 0$. Indeed, if $p(0)$ is even, let $q(n) = p(n) - p(0)$. Given a coloring $\chi : \mathbb{N} \to \{0, \ldots, r\}$, define $\chi'(n) = \chi\left(n + \frac{p(0)}{2}\right)$. A χ'-monochromatic solution $u + v = q(m)$ gives the χ-monochromatic solution $\left(u + \frac{p(0)}{2}\right) + \left(v + \frac{p(0)}{2}\right) = p(m)$.

Khalfalah and Szemerédi proved the conjecture in 2006:

Theorem (Khalfalah–Szemerédi, 2006)

Let $p \in \mathbb{Z}[t]$ satisfy $p(n) \to +\infty$ and $p(0)$ even. Every finite coloring of $\mathbb{N}$ admits distinct monochromatic $x, y \in \mathbb{N}$ and some $n \in \mathbb{N}$ such that

$$x + y = p(n).$$

Hardy field functions

Instead of integer polynomials, we consider the class $\mathcal{B}$ of *Hardy-field functions*, functions $\alpha : (a, \infty) \to \mathbb{R}$ whose germs at $+\infty$ belong to some Hardy field (a family of germs at $+\infty$ which is a field with respect to pointwise sum/product and which is closed under taking derivatives).
$\mathcal{B}$ contains Hardy's field $\mathcal{L}$ of logarithmico-exponential functions.
The field $\mathcal{L}$

- contains the field $\mathbb{R}(t)$ of rational functions;
- is closed under $f \mapsto \exp(f)$, and under $f \mapsto \log|f|$ if $f \neq 0$.
- if $f, g \in \mathcal{L}$, then $f + g, f \cdot g, f' \in \mathcal{L}$, and $1/f \in \mathcal{L}$ when $f \neq 0$;

Examples include $t^{5/2} + t^2 \log t$, $t^3/\log t$, and $t \log\log t$.
Every Hardy-field function f is eventually monotone, so it has a limit:

$$\lim_{x\to\infty} f(x) \in \mathbb{R} \cup \{-\infty, +\infty\}.$$

More generally, if f, g are in a Hardy field, then f/g has a limit in $\mathbb{R} \cup \{\pm\infty\}$

We write $f \prec g$ if $f(x)/g(x) \to 0$ as $x \to \infty$.
We will only consider **sub-polynomials**, i.e., Hardy field functions $\alpha(x)$ such that $\alpha(x) \prec x^l$ for some $l \in \mathbb{N}$. Sub-polynomials have a degree:

$$\deg(\alpha) := \lim_{x\to\infty} \frac{\log(|\alpha(x)|)}{\log(x)} = \sup\{d \in \mathbb{R}; x^d \prec |\alpha(x)|\} \in \mathbb{R} \cup \{-\infty\}.$$

Main coloristic theorem

Let $\alpha(x) \to +\infty$ be a sub-polynomial. The following are equivalent:

1. $\lfloor \alpha(n) \rfloor$ is even for infinitely many $n \in \mathbb{N}$;
2. for every finite coloring of $\mathbb{N}$, there exist distinct monochromatic $x, y \in \mathbb{N}$ and some $n \in \mathbb{N}$ such that

$$x + y = \lfloor \alpha(n) \rfloor.$$

The implication $\neg(1) \Rightarrow \neg(2)$ is immediate: color all sufficiently large integers by parity and give each remaining integer its own color.
Condition (1) can fail only for sub-polynomials of the form

$$\alpha(x) = p(x) + c + r(x), \qquad p \in \mathbb{Q}[t], \quad c \in \mathbb{R}, \quad r(x) \to 0.$$

For "good" sub-polynomials $\alpha(x)$, we can obtain a stronger result. Every sub-polynomial α falls into one of two classes:

1. (**Bad** sub-polynomials) For some $c \in \mathbb{R}$ and $p \in \mathbb{Q}[t]$,

$$\alpha(x) = c\,p(x) + r(x), \qquad |r(x)| \preceq \log x,$$

where r is a Hardy-field function. For example,

$$\sqrt{2}\,x^3 + \sqrt{2}\,x^2 + \log\log x, \qquad x^7, \qquad x^3 - x + 10\log x.$$

2. (**Good** sub-polynomials) For every $c \in \mathbb{R}$ and $p \in \mathbb{Q}[t]$,

$$|\alpha(x) - c\,p(x)| \succ \log x.$$

For example,

$$x^{3/2}, \qquad x^2\log x, \qquad \pi x^2 + x + \log x, \qquad x^3 + x + x^{0.1}.$$

Results of Boshernitzan imply that a sub-polynomial $\alpha(x)$ is good if and only if for all $\kappa > 0$, the sequence $(\kappa\alpha(n))_{n\in\mathbb{N}}$ is uniformly distributed mod 1.

The *upper density* of a set $A \subseteq \mathbb{N}$ is

$$\overline{d}(A) := \limsup_{N\to\infty} \frac{|A \cap [1, N]|}{N}.$$

Density theorem

Let $\alpha(x) \to +\infty$ be a good sub-polynomial.
Then, for every $A \subseteq \mathbb{N}$ with positive upper density, there exist distinct $x, y \in A$ and some $n \in \mathbb{N}$ such that

$$x + y = \lfloor \alpha(n) \rfloor.$$

This theorem implies the coloristic result when $\alpha(x)$ is good. This is because in every finite coloring of $\mathbb{N}$, at least one color class has positive upper density.
We first briefly explain why this result fails for sub-polynomials $\alpha(x) \succ x$ which are not good.
As a first example, let $\alpha(x) = x^2$. The set $A = 3\mathbb{N} + 1$ has positive density but there are no $x, y \in A$ such that $x + y = n^2$, because squares are never 2 modulo 3.

A similar obstruction holds for all integer polynomials p of degree ≥ 2:

Polynomial values are sparse modulo some N

Let $p \in \mathbb{Z}[x]$ have degree at least 2. Then for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\big|\{p(n) \bmod N : n \in \mathbb{Z}\}\big| \leq \varepsilon N.$$

Therefore, given $p \in \mathbb{Z}[x]$ with degree ≥ 2, there is some $N \in \mathbb{N}$ and some residue class $A = k + N\mathbb{Z}$ such that $A + A = 2k + N\mathbb{Z}$ does not contain $p(n)$ for any natural $n \in \mathbb{N}$.
So the density theorem fails for $\alpha = p(x)$.

Every bad sub-polynomial $\alpha(x) \to \infty$ such that $\alpha(x) \succ x$, has the form

$$\alpha(x) = c\,p(x) + r(x), \qquad p \in \mathbb{Z}[x], \quad \deg p \geq 2, \quad |r(x)| \preceq \log x.$$

Using that the residues of p are sparse, one can construct a set A such that $\overline{d}(A) > 0$ but $A + A$ contains no number $\lfloor \alpha(n) \rfloor$ for $n \in \mathbb{N}$.

We will now explain how to deduce the density result from an ergodic theorem.

Unitary ergodic theorem

Let $\alpha(x) \to +\infty$ be a good sub-polynomial. There exists $w\colon \mathbb{N} \to (0, +\infty)$ with $w(N) \searrow 0$ such that, for every Hilbert space $\mathcal{H}$, every unitary map $U\colon \mathcal{H} \to \mathcal{H}$, and every $N \in \mathbb{N}$, we have

$$\left\| \frac{1}{\alpha(N)} \sum_{n \leq N} \alpha'(n) U^{\lfloor \alpha(n) \rfloor} - \frac{1}{\alpha(N)} \sum_{m \leq \lfloor \alpha(N) \rfloor} U^m \right\| \leq w(N).$$

Both expressions inside the norm are "essentially" weighted averages.
Unlike the usual ergodic theorem, this estimate is uniform over all measure-preserving systems.
The proof uses exponential-sum estimates and Herglotz's theorem.
Applying the result to the Koopman operator on $L^2(X, \mu)$ gives the following measure-theoretic version.

Measurable ergodic theorem

Let $\alpha(x) \to +\infty$ be a good sub-polynomial. Then there exists $w\colon \mathbb{N} \to (0, +\infty)$ with $w(N) \searrow 0$ such that, for every invertible probability measure-preserving system $(X, \mathcal{B}, \mu, T)$, every $A, B \in \mathcal{B}$, and every $N \in \mathbb{N}$,

$$\left| \frac{1}{\alpha(N)} \sum_{n \le N} \alpha'(n) \mu\big(T^{\lfloor \alpha(n) \rfloor} A \cap B\big) - \frac{1}{\alpha(N)} \sum_{m \le \lfloor \alpha(N) \rfloor} \mu\big(T^m A \cap B\big) \right| \le w(N).$$

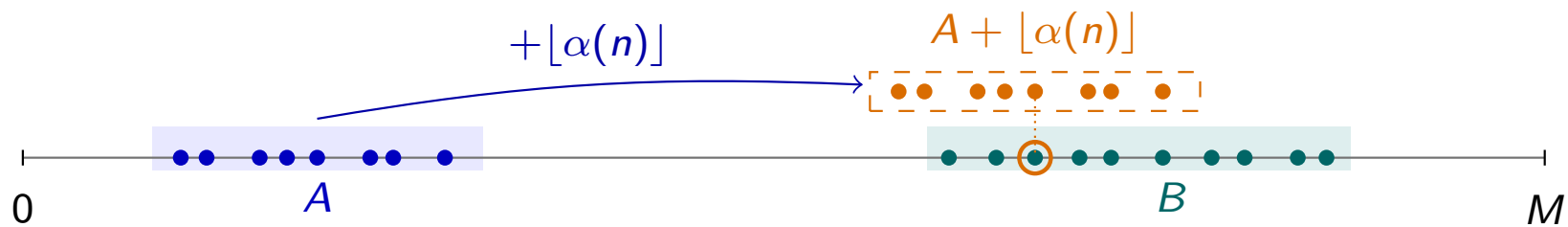
If $\alpha(x) \to +\infty$ is good, we use this theorem to prove the following:

Lemma

Let $\varepsilon > 0$. For all big enough $M \in \mathbb{N}$ and all $A, B \subseteq \{1, \dots, M\}$ such that

$$|A|, |B| \ge \varepsilon M, \qquad \max A < \min B,$$

there exists $n \in \mathbb{N}$ such that $\big(A + \lfloor \alpha(n) \rfloor\big) \cap B \neq \varnothing$.



Sketch of proof of the lemma

We may assume that $M = \lfloor \alpha(N) \rfloor$ for some $N \in \mathbb{N}$.



Let $\overline{A}, \overline{B}$ be the images of A, B under reduction modulo $\lfloor \alpha(N) \rfloor$. We work in the finite measure-preserving system

$$X = \mathbb{Z}/\lfloor \alpha(N) \rfloor \mathbb{Z}, \qquad T : x \mapsto x + 1,$$

with normalized counting measure μ. Then $\mu(\overline{A}), \mu(\overline{B}) \geq \varepsilon$.

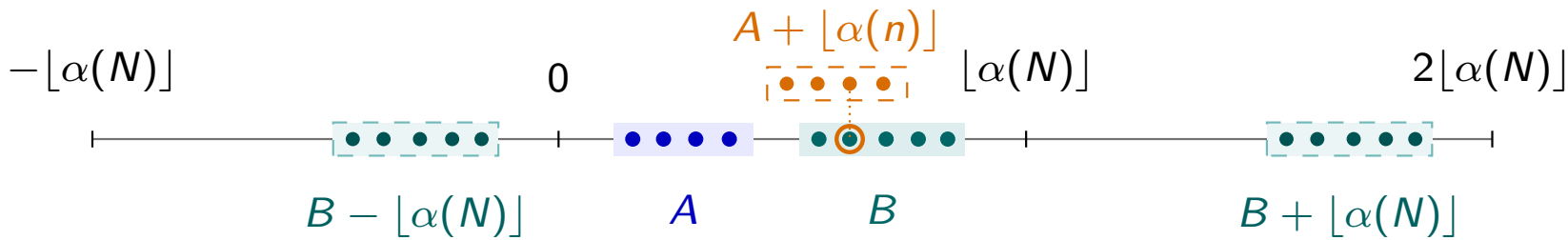
This rotation is ergodic, and averaging over one full cycle gives

$$\frac{1}{\lfloor \alpha(N) \rfloor} \sum_{m=1}^{\lfloor \alpha(N) \rfloor} \mu\big(\overline{A + m} \cap \overline{B}\big) = \mu\big(\overline{A}\big)\, \mu\big(\overline{B}\big) \geq \varepsilon^2.$$

So using our ergodic theorem, if N is large enough we must have

$$\frac{1}{\alpha(N)} \sum_{n=1}^{N} \alpha'(n) \mu\Big(\overline{A + \lfloor \alpha(n) \rfloor} \cap \overline{B}\Big) \geq \frac{\varepsilon^2}{2}.$$

In particular, $\overline{A + \lfloor \alpha(n) \rfloor} \cap \overline{B} \neq \varnothing$ for some $n \leq N$. As $\alpha(n) \leq \alpha(N)$, this implies $\big(A + \lfloor \alpha(n) \rfloor\big) \cap B \neq \varnothing$, because $A + \lfloor \alpha(n) \rfloor$ cannot intersect any translate $B + j\lfloor \alpha(N) \rfloor$, $j \in \mathbb{Z} \setminus \{0\}$. □



We considered $A, B \subseteq \{1, \ldots, M\}$, but the same argument works when A and B lie in any interval of length M, $\{k+1, \ldots, k+M\}$.

The conclusion we obtain is much stronger than $\big(A + \lfloor \alpha(n) \rfloor\big) \cap B \neq \varnothing$; we have $\big|\big(A + \lfloor \alpha(n) \rfloor\big) \cap B\big| > \frac{\varepsilon^2}{3} M$ for "many" $n \in \mathbb{N}$.

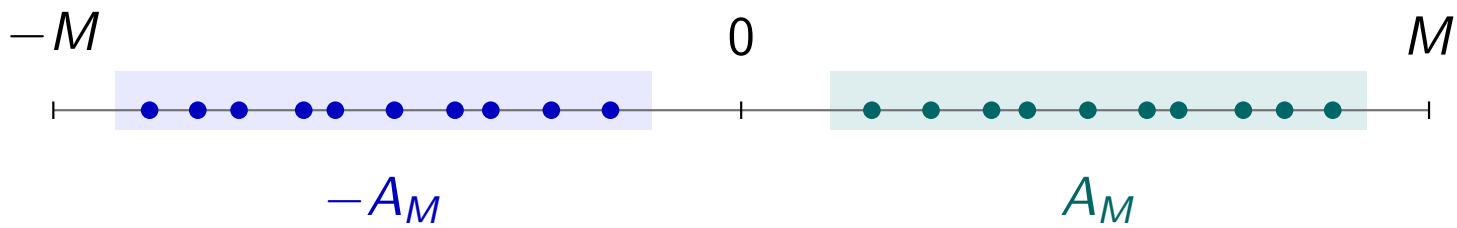
Corollary

If $A \subseteq \mathbb{N}$ and $\overline{d}(A) > 0$, then there exist $x, y \in A$ and $n \in \mathbb{N}$ such that

$$x + y = \lfloor \alpha(n) \rfloor.$$

Proof. Let $0 < \varepsilon < \overline{d}(A)$. There are arbitrarily large $M \in \mathbb{N}$ for which

$$A_M := A \cap \{1, \ldots, M\} \qquad \text{satisfies} \qquad |A_M| > \varepsilon M.$$



Therefore, for large enough M, the previous lemma gives some $n \in \mathbb{N}$ such that

$$\big(\lfloor \alpha(n) \rfloor - A_M\big) \cap A_M \neq \varnothing.$$

Thus $\lfloor \alpha(n) \rfloor \in A_M + A_M \subseteq A + A$. □

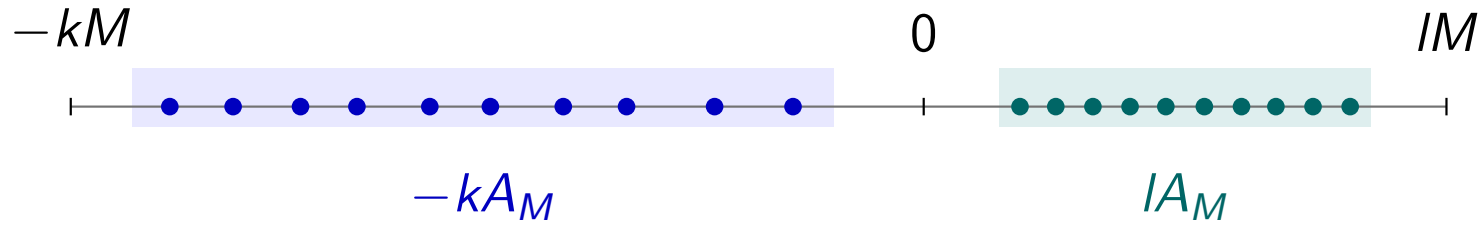
In fact, we have $\big|\big(\lfloor \alpha(n) \rfloor - A_M\big) \cap A_M\big| \geq 2$ for some M, so the elements $x, y \in A$ such that $x + y = \lfloor \alpha(n) \rfloor$ can be taken to be distinct.

More general corollary

If $k, l \in \mathbb{N}$, $A \subseteq \mathbb{N}$ and $\overline{d}(A) > 0$, then there exist $x, y \in A$ and $n \in \mathbb{N}$ such that

$$kx + ly = \lfloor \alpha(n) \rfloor.$$

Proof. Set $A_M = A \cap \{1, \ldots, M\}$. For arbitrarily large M, there is $n \in \mathbb{N}$ such that $\left(\lfloor \alpha(n) \rfloor - kA_M\right) \cap lA_M \neq \varnothing$.



Thus $\lfloor \alpha(n) \rfloor \in kA_M + lA_M \subseteq kA + lA$. □

A similar argument also proves one can solve the equation

$$kx - ly = \lfloor \alpha(n) \rfloor,$$

when $k, l \in \mathbb{N}$, $k \geq l$ (if $k < l$ and $\alpha(x) \succ x$, the corresponding result does not hold).

For $A, B \subseteq \mathbb{N}$, define their *joint upper density* by

$$\overline{d}(A,B) := \limsup_{N\to\infty} \min\left\{ \frac{|A \cap \{1,\dots,N\}|}{N}, \frac{|B \cap \{1,\dots,N\}|}{N} \right\}.$$

Thus $\overline{d}(A,B) > 0$ means that, along arbitrarily large values of N, both A and B occupy a fixed positive proportion of $\{1,\dots,N\}$.

Two-set density theorem

Let $k, l \in \mathbb{N}$. If $A, B \subseteq \mathbb{N}$ satisfy $\overline{d}(A,B) > 0$, then there exist $x \in A$, $y \in B$, and $n \in \mathbb{N}$ such that

$$kx + ly = \lfloor \alpha(n) \rfloor.$$

Two-set density theorem

Let $k, l \in \mathbb{N}$. If $A, B \subseteq \mathbb{N}$ satisfy $\overline{d}(A, B) > 0$, then there exist $x \in A$, $y \in B$, and $n \in \mathbb{N}$ such that

$$kx + ly = \lfloor \alpha(n) \rfloor. \tag{1}$$

We can also ask $kx + ly$ to be a 'good approximation' for $\alpha(n)$: applying (1) to $10\alpha(n), 10A, 10B$, we obtain the following sharpening:

Good approximation

Let $k, l \in \mathbb{N}$. If $A, B \subseteq \mathbb{N}$ satisfy $\overline{d}(A, B) > 0$, then there exist $x \in A$, $y \in B$, and $n \in \mathbb{N}$ such that

$$\alpha(n) - (kx + ly) \in [0, 0.1).$$

We can also replace the floor function $\lfloor \cdot \rfloor$ by the nearest-integer function $[\cdot]$ or the ceiling function $\lceil \cdot \rceil$, because if $\alpha(x)$ is good, then $\alpha(x) + c$ is good for all $c \in \mathbb{R}$.

What about bad sub-polynomials?

Suppose $\alpha(x)$ is bad,

$$\alpha(x) = c\,p(x) + r(x), \qquad c \in \mathbb{R}, \quad p \in \mathbb{Q}[x], \quad |r(x)| \preceq \log x.$$

If $\lfloor \alpha(n) \rfloor$ is even for infinitely many $n \in \mathbb{N}$, we still have a 'density result':

Finitary density lemma

For every $\varepsilon > 0$, there exists a finite set $F = F(\varepsilon, \alpha) \subseteq \mathbb{N}$ such that, whenever $A, B \subseteq F$ satisfy

$$|A|, |B| \geq \varepsilon |F|,$$

there exist distinct $x \in A$, $y \in B$, and $n \in \mathbb{N}$ such that

$$x + y = \lfloor \alpha(n) \rfloor.$$

If $c = 1$, we can let F be a finite arithmetic progression.



Proof of the coloristic theorem for bad sub-polynomials

Suppose $\alpha(x)$ is a bad sub-polynomial, $\lfloor \alpha(n) \rfloor$ is even for infinitely many n, and

$$\mathbb{N} = C_1 \cup \cdots \cup C_q$$

is a finite coloring. By the previous lemma, there is a finite set $F \subseteq \mathbb{N}$ such that for all $A \subseteq F$ satisfying

$$|A| \geq \frac{|F|}{q},$$

there are distinct $x, y \in A$ such that $x + y = \lfloor \alpha(n) \rfloor$ for some $n \in \mathbb{N}$. By the pigeonhole principle, some color class C_i satisfies

$$|C_i \cap F| \geq \frac{|F|}{q},$$

so there must be distinct $x, y \in C_i$ such that $x + y = \lfloor \alpha(n) \rfloor$ for some $n \in \mathbb{N}$. $\square$

Quantitative results

Back to good sub-polynomials.
We recall the ergodic theorem, and in particular the function $w(N)$:

Ergodic theorem

Let $\alpha(x) \to +\infty$ be a good sub-polynomial. Then there exists $w\colon \mathbb{N} \to (0, +\infty)$ with $w(N) \searrow 0$ such that, for every invertible probability measure-preserving system $(X, \mathcal{B}, \mu, T)$, every $A, B \in \mathcal{B}$, and every $N \in \mathbb{N}$,

$$\left| \frac{1}{\alpha(N)} \sum_{n \leq N} \alpha'(n) \mu\big(T^{\lfloor \alpha(n) \rfloor} A \cap B\big) - \frac{1}{\alpha(N)} \sum_{m \leq \lfloor \alpha(N) \rfloor} \mu\big(T^m A \cap B\big) \right| \leq w(N).$$

Quantitative upper bounds on $w(N)$ can be used to obtain sharper results. Our bounds on $w(N)$ depend on what kind of good sub-polynomial we are dealing with.

Decomposition of sub-polynomials

Any sub-polynomial $\alpha(x)$ satisfies one of the following two conditions:

- α is **special**, meaning that

$$\lim_{x\to\infty} \frac{|\alpha(x)|}{x^l} \in \{0, \infty\} \text{ for all } l = 1, 2, \ldots.$$

  E.g. $x^\pi, x \log\log(x)$.

- α can be expressed uniquely as

$$\alpha(x) = a_d x^d + \cdots + a_j x^j + r_\alpha(x),$$

  where $d \geq j \geq 1$ are natural numbers, $a_i \in \mathbb{R}$ for all i $a_d, a_j \neq 0$, $|r_\alpha(x)| \prec x^j$ and $r_\alpha(x)$ is special.

We call r_α the special part of α, and let $r_\alpha = \alpha$ if α is special.

A sub-polynomial α is good when either $|r_\alpha(x)| \succ \log(x)$, or the polynomial part $a_d x^d + \cdots + a_j x^j$ is not in $\bigcup_{c\in\mathbb{R}} c\mathbb{Q}[x]$.

Good sub-polynomials far from all polynomials

We first consider good sub-polynomials α which satisfy $|r_\alpha(x)| \succ \log(x)$.

E.g. $x^\pi, \frac{x^3}{\log(x)}, x^2 + x + \log(x) \cdot \log(\log(x))$.

In this case, if $\alpha(x) \succeq x$ and $d = \deg(\alpha)$, for big enough N we can let

$$w(N) = \left(\frac{\log(N)}{|r_\alpha(N)|}\right)^{\frac{1}{d \cdot 2^{d+8}}}.$$

For example, if $\deg(r_\alpha) > 0$, then we can let $w(N) = \frac{1}{N^\delta}$ for some $\delta > 0$.
Here are some nice combinatorial corollaries using how fast $w(N)$ approaches zero:

Sums of sets of zero density

Theorem

Let $\alpha(x) \to \infty$ be a sub-polynomial such that $\deg(r_\alpha) > 0$. Then there is $\delta > 0$ such that, for all sufficiently large N and all $A, B \subseteq \{1, \dots, N\}$ such that

$$|A|, |B| \geq \frac{N}{N^\delta},$$

there are $x \in A, y \in B$ such that $x + y = \lfloor \alpha(n) \rfloor$ for some $n \in \mathbb{N}$.

Large subsets of the primes

Let $\mathcal{P}$ be the set of prime numbers. For $A \subseteq \mathcal{P}$, define its *relative upper density* by

$$\overline{d}_{\mathcal{P}}(A) := \limsup_{N\to\infty} \frac{|A \cap \{1,\dots,N\}|}{|\mathcal{P} \cap \{1,\dots,N\}|}.$$

Theorem

Let $\alpha(x) \to \infty$ be a sub-polynomial such that $|r_\alpha(x)| \succ (\log x)^l$ for every $l > 0$. Then, for every $A \subseteq \mathcal{P}$ with $\overline{d}_{\mathcal{P}}(A) > 0$, there exist arbitrarily large $x, y \in A$ and some $n \in \mathbb{N}$ such that

$$x + y = \lfloor \alpha(n) \rfloor.$$

Good sub-polynomials close to some polynomial

Finally, we consider good sub-polynomials α of the form

$$\alpha(x) = a_d x^d + \cdots + a_1 x + r_\alpha(x), \quad |r_\alpha(x)| \preceq \log x,$$

where $a_i \in \mathbb{R}$, $a_d > 0$.

We cannot have $\frac{a_j}{a_d} \in \mathbb{Q}$ for all $1 \leq j < d$; then α would be bad and the ergodic theorem would be false (i.e. we cannot have $w(N) \to 0$).

E.g. $x^3 + \pi x$, or $\sqrt{2}x^4 + \pi x^2 + \log(x + \log(x))$.

Letting $j_0 \in \mathbb{N}$ be the largest exponent such that $\frac{a_{j_0}}{a_d} \notin \mathbb{Q}$, our upper bound on $w(N)$ depends on how well $\frac{a_{j_0}}{a_d}$ can be approximated by rationals.
In particular, if $\frac{a_{j_0}}{a_d}$ has finite irrationality exponent, then we can let $w(N)$ be $N^{-\delta}$ for some $\delta > 0$.

The end

Questions?